

Topic 91 - Practice Assignment: Connecting and Transforming Data with Power BI Desktop

Using your Adventure Works Power BI file, complete the following:

1) Create new queries to connect to the **Product_Categories** and **Product_Subcategories** files from the course resources:

- Name your queries **Product_Category_Lookup** and **Product_Subcategory_Lookup**
- Confirm that headers have been promoted and that detected data types are correct
- Disable the report refresh option for both connections

2) Make the following modifications to the **Product_Lookup** query:

- Add a calculated column that extracts all characters before the dash ("-") in the **ProductSKU** column, named "**SKUType**"
- Update the **SKUType** calculation above to return all characters before *second* dash, instead of the first
- Replace zeros (0) in the **ProductStyle** column with "NA"
- Update the **DiscountPrice** calculation to 15%, by multiplying the **ProductPrice** values by 0.85 (*instead of 0.9*)

3) Using the Statistics tools in the Query Editor, confirm the following values:

- Average product cost (**\$413.66**)
- Number of distinct product colors (**10**)
- Number of distinct customer names (**18,110**)
- Maximum annual customer income (**\$170,000**)
- Count of order numbers (**56,046**)
- Count of distinct order numbers (**25,164**)

4) Make the following modifications to the **Customer_Lookup** query:

- Add a new calculated column for the month of birth (named "**Birth Month**"), based on **BirthDate**
- Add a calculated column (named "**Birth Month-Year**") to merge **Birth Month** with **Birth Year**
- Add a conditional column to categorize customer income (named "**IncomeLevel**"), based on the following criteria:
 - If **AnnualIncome** >= \$150,000, then **IncomeLevel** = "Very High"
 - If **AnnualIncome** >= \$100,000, then **IncomeLevel** = "High"
 - If **AnnualIncome** >= \$50,000, then **IncomeLevel** = "Average"
 - Otherwise **IncomeLevel** = "Low"

5) Apply all changes, and confirm that new tables and fields are accessible in the **Data** view. Save the Power BI file.